

Figure 1: Conventional understanding of AI [Image credit: Geeky-Gadgets]

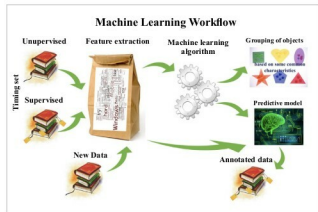


Figure 2: Machine learning workflow [Image credit: TomBone's Computer Vision Blog]

While it began as a small part of AI, burgeoning interest has propelled ML to the forefront of research and it is now used across domains. Growing hardware support as well as improvements in algorithms, especially pattern recognition, has led to ML being accessible for a much larger audience, leading to wider adoption.

Applications of ML

Initially, the primary applications of ML were limited to the field of computer vision and pattern recognition. This was prior to the stellar success and accuracy it enjoys today. Back then, ML seemed a pretty tame field, with its scope limited to education and academics.

Today we use ML without even being aware of how dependent we are on it for our daily activities. From Google's search team trying to replace the PageRank algorithm with an improved ML algorithm named RankBrain, to Facebook automatically suggesting friends to tag in a picture, we are surrounded by use cases for ML algorithms.

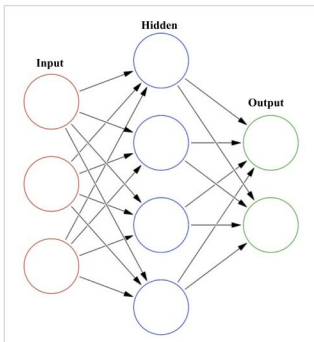


Figure 3: Artificial neural networks [Image credit: Shutterstock]

Deep learning (DL)

A key ML approach that remained dormant for a few decades was artificial neural networks. This eventually gained wide acceptance when improved processing capabilities became available. A neural network simulates the activity of a brain's neurons in a layered fashion, and the propagation of data occurs in a similar manner, enabling machines to learn more about a given set of observations and make accurate predictions.

These neural networks that had until recently been ignored, save for a few researchers led by Geoffrey Hinton, have today demonstrated an exceptional potential for handling large volumes of data and enhancing the practical applications of machine learning. The accuracy of these models allows reliable services to be offered to end users, since the false positives have been eliminated almost entirely.

Applications of DL

DL has large scale business applications because of its capacity to learn from millions of observations at once. Although computationally intensive, it is still the preferred alternative because of its unparalleled accuracy. This encompasses a number of image recognition applications that conventionally relied on computer vision practices until the emergence of DL. Autonomous vehicles and recommendation systems (such as those used by Netflix and Amazon) are among the most popular applications of DL algorithms.

Comparing AI, ML and DL

Comparing the techniques: The term AI was defined in the Dartmouth Conference (1956) as follows: "Every aspect of learning or any other feature of intelligence can in principle